UMBC Workout Manager

- **Background:**
  - A daily workout regimen is a cornerstone of maintaining good health and well-being. Regular exercise offers a plethora of physical and mental benefits. It not only helps you build and maintain a healthy weight but also strengthens your muscles and bones, enhances cardiovascular health, and improves flexibility and balance. Engaging in a daily workout routine can boost your energy levels, reducing fatigue and enhancing your overall stamina. Exercise also releases endorphins, which are natural mood boosters, helping to combat stress, anxiety, and depression.
  - Beyond the immediate physical benefits, a daily workout plays a critical role in disease prevention. Regular physical activity can reduce the risk of chronic conditions such as heart disease, diabetes, and high blood pressure. It helps regulate blood sugar levels, improves insulin sensitivity, and promotes healthy cholesterol levels. Furthermore, exercise contributes to a more robust immune system, making the body better equipped to fend off infections and illnesses. Overall, a daily workout is a proactive step toward a longer, healthier life, as it not only enhances your physical health but also bolsters mental clarity and emotional well-being, fostering a higher quality of life.
  - For this project, we are going to design a tool to manage workouts. For example, a workout might include a variety of different exercises including some that work the arms, legs, or back.
- **Assignment Description:**
  - Your assignment is to build an application that will allow the user to create and manage a variety of workouts.
- Project Structure
  - Workout-Manager/
  - │
  - ├── Exercise.cpp
  - ├── Exercise.h
  - ├── Workout.cpp
  - ├── Workout.h
  - ├── Manager.cpp
  - ├── Manager.h
  - ├── proj3.cpp
  - ├── makefile
  - |------ proj3_ta1.txt
  - |____ proj3 ta2.txt
  - |____ proj3 ta3.txt
  - ├── proj3_workout1.txt
  - ├── proj3_workout2.txt
  - └── proj3_workout3.txt
  -

- **All the files for Project3 Workout Manager:**

```
[kpatel38@linux3 project3]$ ls
Exercise.cpp  Manager.o          proj3_ta2.txt       Workout.cpp
Exercise.h    proj3              proj3_ta3.txt       Workout.h
Exercise.o    proj3.cpp          proj3_workout1.txt  Workout.o
makefile      proj3_sample1.txt  proj3_workout2.txt
Manager.cpp   proj3_sample2.txt  proj3_workout3.txt
Manager.h     proj3_ta1.txt      vgcore.1374357
[kpatel38@linux3 project3]$
```

- **Sample Input and Output**
    - **To run file type:**
        - ./proj3 proj3_workout1.txt

```
[kpatel38@linux3 project3]$ ./proj3 proj3_workout1.txt

***Manager***

Opened File
5 exercises loaded across 1 workouts.
What would you like to do?:
1. Display Workouts
2. Reverse Workouts
3. Insert New Exercise
4. Exit
1
******************
Workout for Monday
5 exercises
Back:Barbell Shrug(3@60lbs)
Chest:Bench Press(3@205lbs)
Legs:Toe Raises(3@0lbs)
Back:Deadlift(4@115lbs)
Legs:Seated Leg Press(4@145lbs)


What would you like to do?:
1. Display Workouts
2. Reverse Workouts
3. Insert New Exercise
4. Exit
2
Done reversing Workout.
What would you like to do?:
1. Display Workouts
2. Reverse Workouts
3. Insert New Exercise
4. Exit
1
******************
Workout for Monday
5 exercises
Legs:Seated Leg Press(4@145lbs)
Back:Deadlift(4@115lbs)
Legs:Toe Raises(3@0lbs)
Chest:Bench Press(3@205lbs)
Back:Barbell Shrug(3@60lbs)
```

```
What would you like to do?:
1. Display Workouts
2. Reverse Workouts
3. Insert New Exercise
4. Exit
3
What is the name of the workout you would like to insert into?
Monday
What is the category of the exercise?
Legs
What is the name of the exercise?
Squats
How many reps did you do?
3
How much weight did you use? (in lbs.)
0
What would you like to do?:
1. Display Workouts
2. Reverse Workouts
3. Insert New Exercise
4. Exit
1
******************
Workout for Monday
6 exercises
Legs:Seated Leg Press(4@145lbs)
Back:Deadlift(4@115lbs)
Legs:Toe Raises(3@0lbs)
Chest:Bench Press(3@205lbs)
Back:Barbell Shrug(3@60lbs)
Legs:Squats(3@0lbs)


What would you like to do?:
1. Display Workouts
2. Reverse Workouts
3. Insert New Exercise
4. Exit
4
Workouts removed from memory
Deleting Workouts
[kpatel38@linux3 project3]$
```

- Here is a longer example run where there are 7 different workouts (one for

- each day).
- **To run the file type:**
  - ./proj3 proj3_workout2.txt

```
[kpatel38@linux3 project3]$ ./proj3 proj3_workout2.txt

***Manager***

Opened File
50 exercises loaded across 7 workouts.
What would you like to do?:
1. Display Workouts
2. Reverse Workouts
3. Insert New Exercise
4. Exit
1
******************
Workout for Tuesday
6 exercises
Shoulder:Dumbbell Horizontal External Shoulder Rotation(5@160lbs)
Back:One-Handed Lat Pulldown(10@15lbs)
Leg:Hack Squat Machine(7@60lbs)
Abs:Mountain Climbers(10@115lbs)
Tricep:Bench Dip(10@0lbs)
Bicep:Cable Curl With Bar(8@155lbs)


Workout for Monday
5 exercises
Glute:Clamshells(4@35lbs)
Back:Back Extension(4@35lbs)
Glute:Glute Bridge(10@40lbs)
Back:Snatch(7@230lbs)
Leg:Shallow Body Weight Lunge(8@135lbs)


Workout for Sunday
6 exercises
Glute:Dumbbell Frog Pumps(5@65lbs)
Chest:Resistance Band Chest Fly(8@0lbs)
Back:Lat Pulldown With Supinated Grip(9@185lbs)
Chest:Floor Press(5@140lbs)
Bicep:Dumbbell Curl(5@35lbs)
Back:Barbell Shrug(5@125lbs)


Workout for Thursday
7 exercises
Leg:Body Weight Lunge(5@45lbs)
Forearm:Plate Wrist Curl(10@20lbs)
Shoulder:Dumbbell Horizontal External Shoulder Rotation(7@30lbs)
Shoulder:Reverse Machine Fly(4@20lbs)
Shoulder:Barbell Upright Row(7@45lbs)
Chest:Incline Dumbbell Press(2@35lbs)
Leg:Barbell Lunge(5@30lbs)
```

```
Workout for Friday
7 exercises
Shoulder:Snatch Grip Behind the Neck Press(2@85lbs)
Abs:Lying Windshield Wiper(9@35lbs)
Bicep:Incline Dumbbell Curl(10@80lbs)
Abs:Kneeling Side Plank(3@0lbs)
Leg:Chair Squat(2@100lbs)
Forearm:Farmers Walk(1@90lbs)
Back:Pause Deadlift(10@215lbs)


Workout for Wednesday
9 exercises
Bicep:Bodyweight Curl(2@10lbs)
Chest:Resistance Band Chest Fly(3@0lbs)
Chest:Bench Press(7@230lbs)
Leg:Lying Leg Curl(3@95lbs)
Abs:Plank(4@0lbs)
Back:Inverted Row(9@75lbs)
Bicep:Machine Bicep Curl(10@80lbs)
Shoulder:Machine Shoulder Press(7@75lbs)
Back:Back Extension(7@10lbs)


Workout for Saturday
10 exercises
Shoulder:Reverse Dumbbell Flyes(3@15lbs)
Abs:Lying Windshield Wiper with Bent Knees(10@10lbs)
Shoulder:Seated Smith Machine Shoulder Press(8@190lbs)
Abs:Hanging Knee Raise(9@20lbs)
Abs:Side Plank(2@0lbs)
Shoulder:Bench Press(9@160lbs)
Shoulder:Plate Front Raise(5@10lbs)
Leg:Barbell Walking Lunge(7@30lbs)
Shoulder:Band External Shoulder Rotation(4@135lbs)
Back:Good Morning(2@20lbs)


What would you like to do?:
1. Display Workouts
2. Reverse Workouts
3. Insert New Exercise
4. Exit
```

- One way to test to make sure that you have successfully removed any of the memory leaks is to use the valgrind command.
- Since this project makes extensive use of dynamic memory, it is important that you test your program for memory leaks using valgrind: valgrind ./proj3 proj3_workout1.txt
- If you have no memory leaks, you should see output like the following:

```
==4137800== HEAP SUMMARY:
==4137800==     in use at exit: 0 bytes in 0 blocks
==4137800==   total heap usage: 17 allocs, 17 frees, 84,009 bytes allocated
==4137800==
==4137800== All heap blocks were freed -- no leaks are possible
==4137800==
==4137800== For lists of detected and suppressed errors, rerun with: -s
==4137800== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```